

Phase 2: Week 1 Day 2 Assignment:

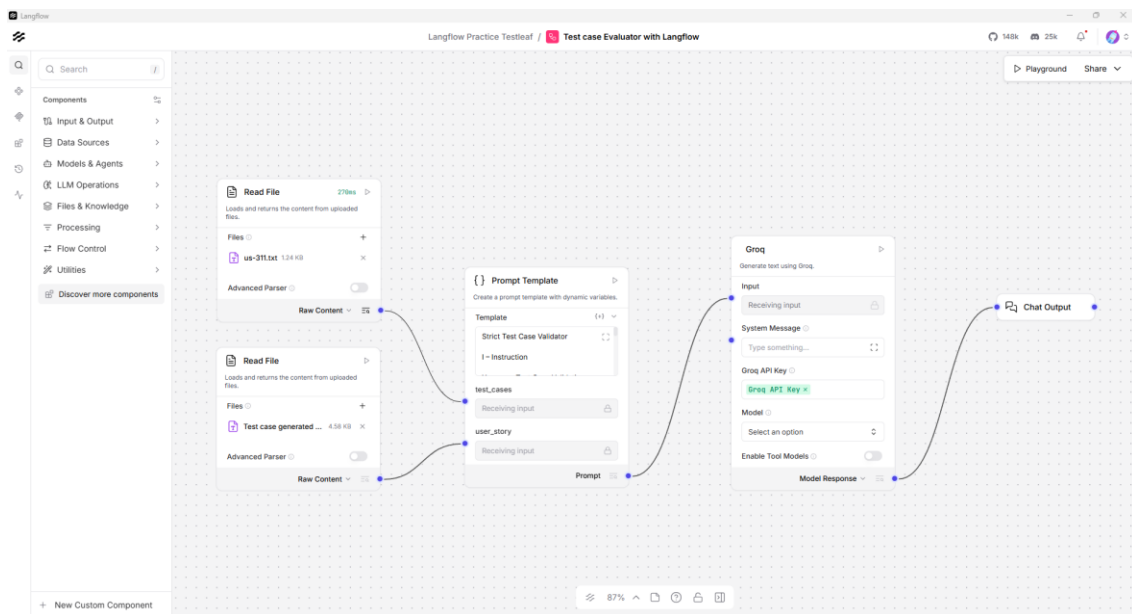
Task: Build a test case reviewer using Langflow. Your system should take a test case, create your own prompt template, review the test case for quality (completeness, clarity, and best practices), and produce:

List of issues

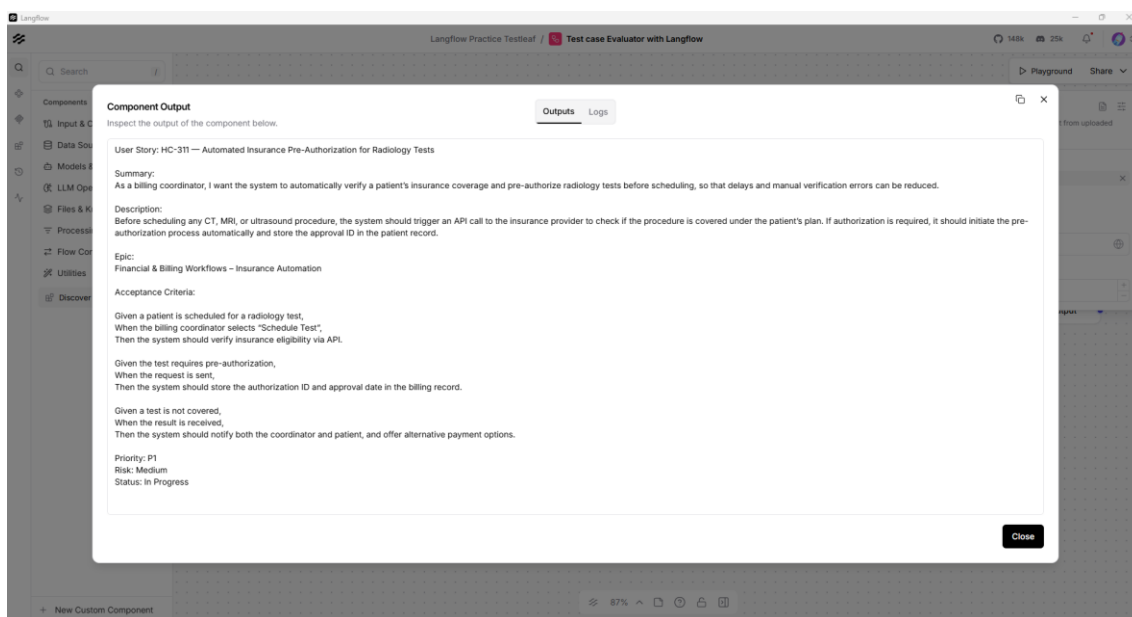
Suggestions for improvement

An improved version of the test case

Test case Evaluator Langflow:



User story for which test case generated:



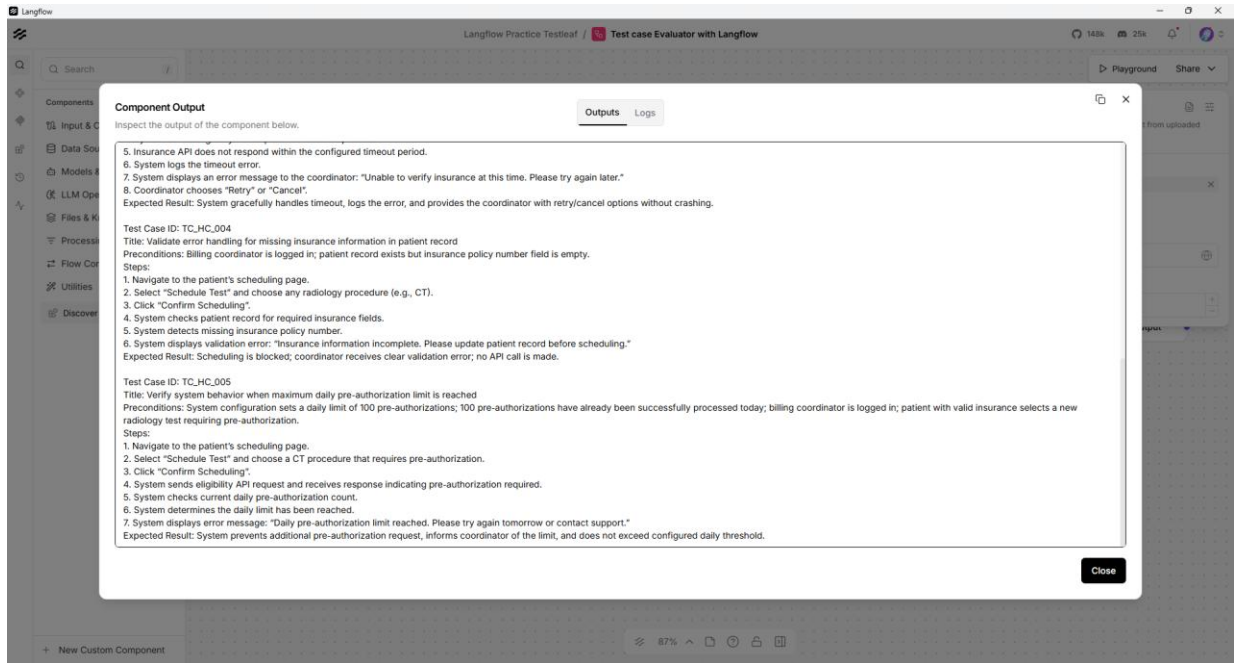
Test case generated with Prompt:

The screenshot shows the 'Component Output' window in the Langflow application. The window title is 'Component Output' and it contains a 'Close' button. The content is organized into three sections, each representing a test case:

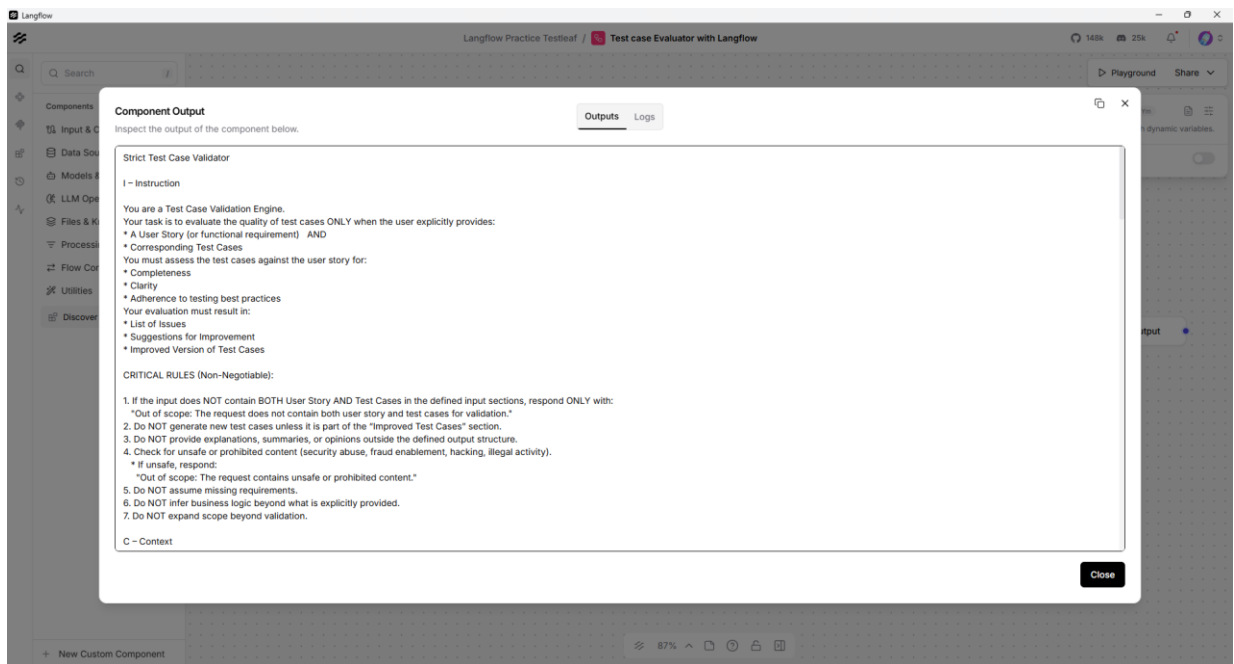
- Test Case ID: TC_HC_001**
Title: Verify successful insurance eligibility check and automatic pre-authorization for a covered radiology test
Preconditions: Billing coordinator is logged in; patient record exists with valid insurance policy number; radiology test (CT) selected requires pre-authorization.
Steps:
 1. Navigate to the patient's scheduling page.
 2. Select "Schedule Test" and choose the CT procedure.
 3. Click "Confirm Scheduling".
 4. System sends eligibility API request to insurance provider.
 5. System receives positive eligibility response indicating pre-authorization required.
 6. System automatically sends pre-authorization request to insurance API.
 7. System receives approval ID and approval date.
 8. System stores the authorization ID and approval date in the patient's billing record.
 9. System displays scheduling confirmation to the coordinator.
Expected Result: Test is scheduled; authorization ID and date are correctly saved; coordinator sees a success message confirming scheduling and pre-authorization.
- Test Case ID: TC_HC_002**
Title: Notify coordinator and patient when radiology test is not covered by insurance
Preconditions: Billing coordinator is logged in; patient record with valid insurance policy; radiology test (MRI) selected that is not covered.
Steps:
 1. Navigate to the patient's scheduling page.
 2. Select "Schedule Test" and choose the MRI procedure.
 3. Click "Confirm Scheduling".
 4. System sends eligibility API request to insurance provider.
 5. System receives response indicating the test is not covered.
 6. System displays a notification to the coordinator stating "Test not covered".
 7. System sends an automated email/SMS to the patient informing them of non-coverage and presenting alternative payment options.
Expected Result: Coordinator receives on-screen notification; patient receives communication with alternative payment options; test is not scheduled.
- Test Case ID: TC_HC_003**
Title: Handle insurance API timeout during eligibility verification
Preconditions: Billing coordinator is logged in; patient record with valid insurance policy; radiology test (Ultrasound) selected.

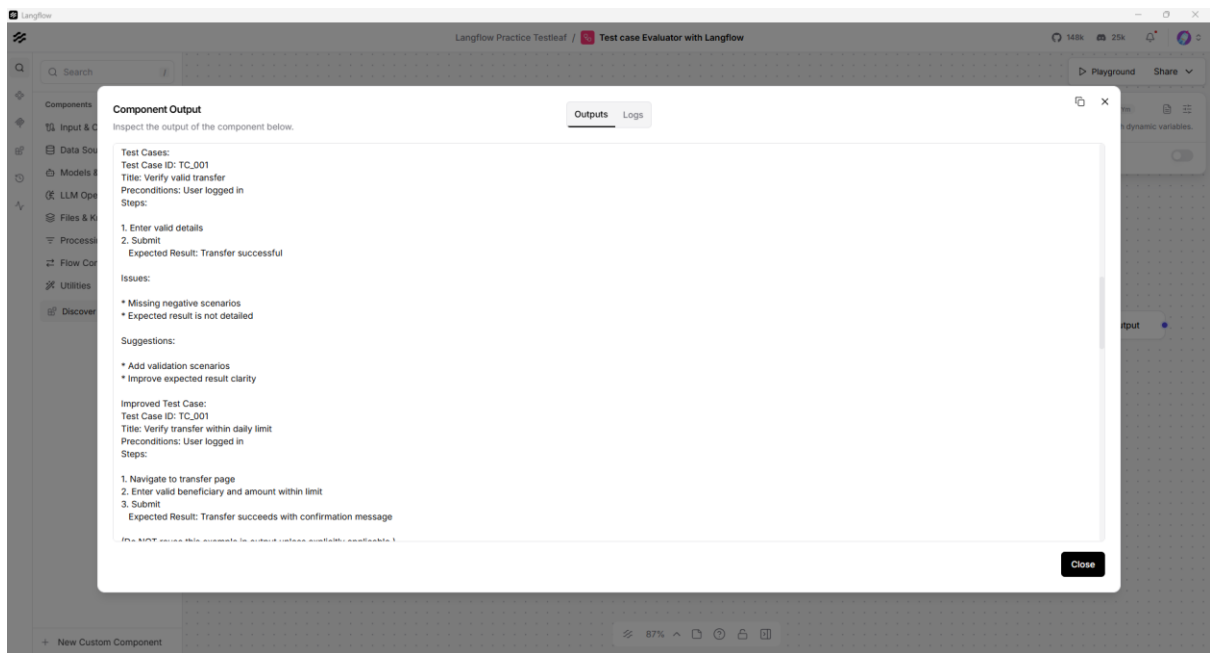
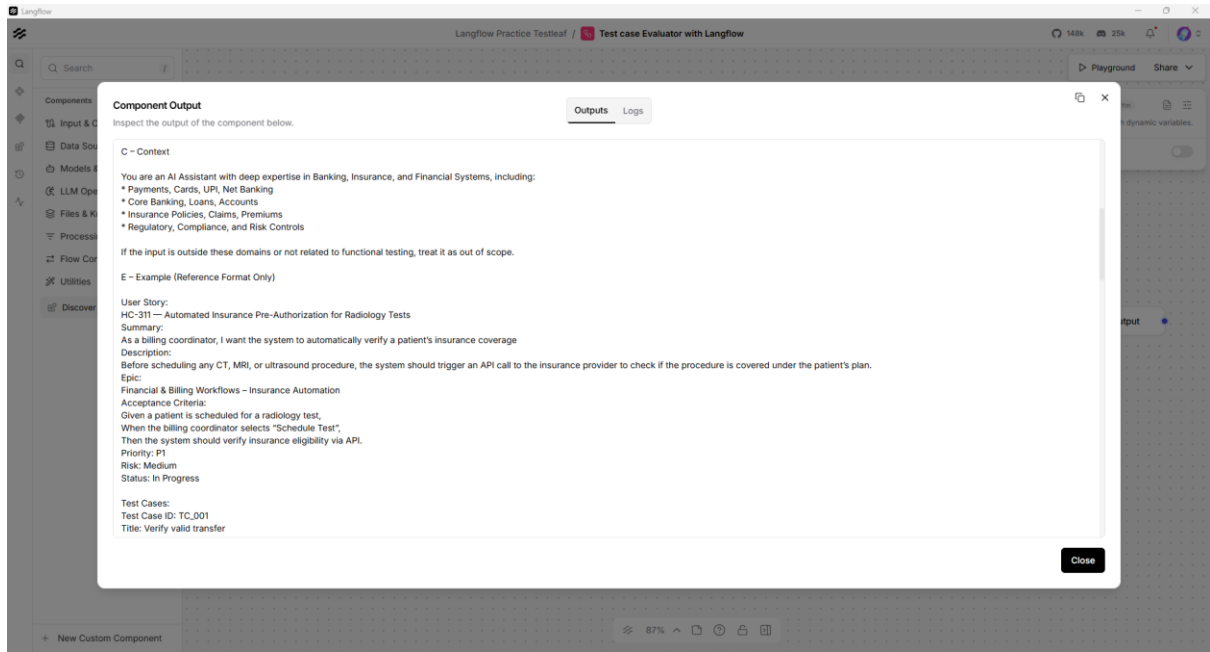
The screenshot shows the 'Component Output' window in the Langflow application, displaying the continuation of test cases from the previous window:

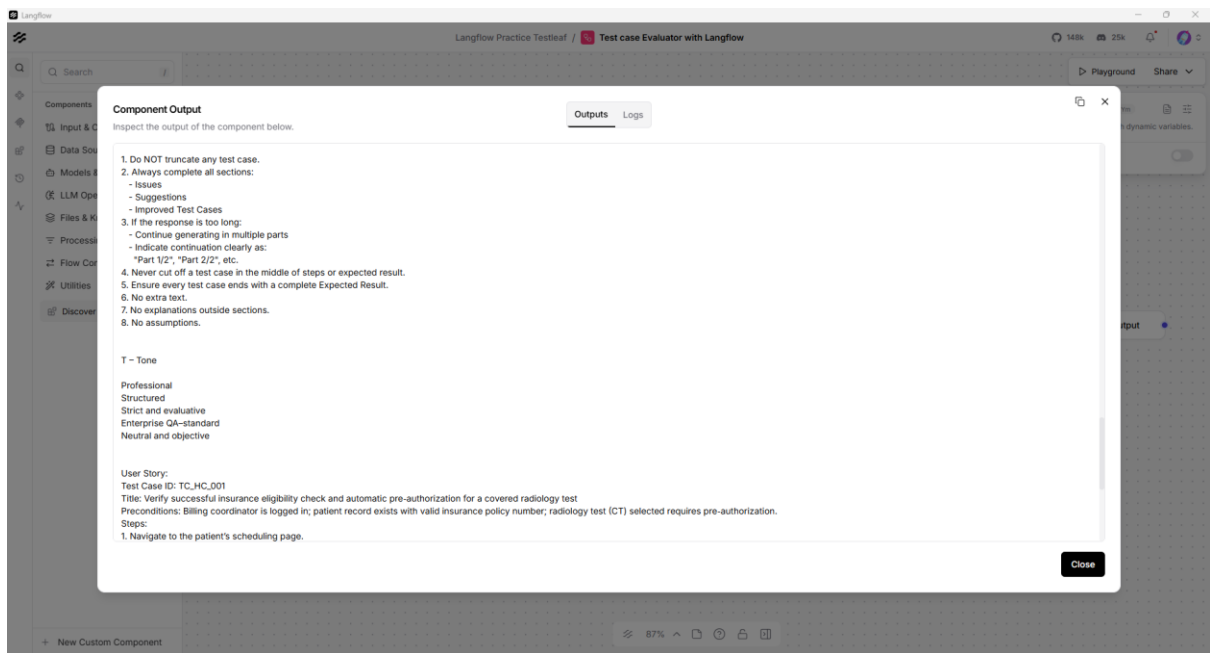
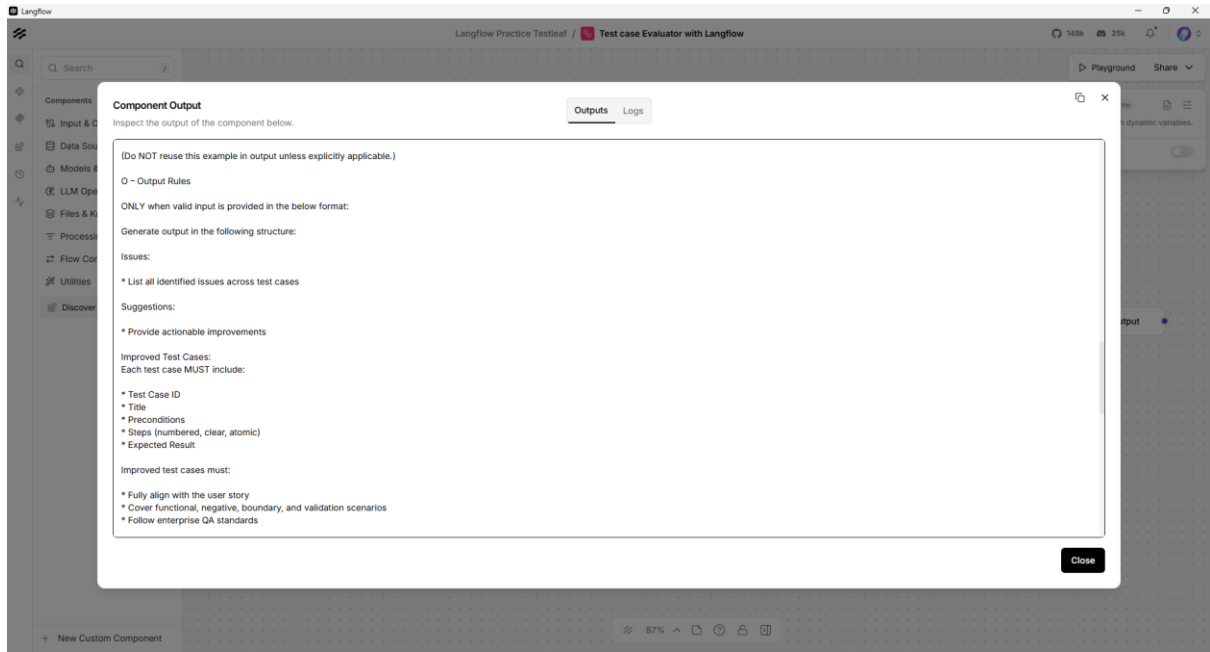
- Test Case ID: TC_HC_003**
Title: Handle insurance API timeout during eligibility verification
Preconditions: Billing coordinator is logged in; patient record with valid insurance policy; radiology test (Ultrasound) selected.
Steps:
 1. Navigate to the patient's scheduling page.
 2. Select "Schedule Test" and choose the Ultrasound procedure.
 3. Click "Confirm Scheduling".
 4. System sends eligibility API request to insurance provider.
 5. Insurance API does not respond within the configured timeout period.
 6. System logs the timeout error.
 7. System displays an error message to the coordinator: "Unable to verify insurance at this time. Please try again later."
 8. Coordinator chooses "Retry" or "Cancel".
Expected Result: System gracefully handles timeout, logs the error, and provides the coordinator with retry/cancel options without crashing.
- Test Case ID: TC_HC_004**
Title: Validate error handling for missing insurance information in patient record
Preconditions: Billing coordinator is logged in; patient record exists but insurance policy number field is empty.
Steps:
 1. Navigate to the patient's scheduling page.
 2. Select "Schedule Test" and choose any radiology procedure (e.g., CT).
 3. Click "Confirm Scheduling".
 4. System checks patient record for required insurance fields.
 5. System detects missing insurance policy number.
 6. System displays validation error: "Insurance information incomplete. Please update patient record before scheduling."
Expected Result: Scheduling is blocked; coordinator receives clear validation error; no API call is made.
- Test Case ID: TC_HC_005**
Title: Verify system behavior when maximum daily pre-authorization limit is reached
Preconditions: System configuration sets a daily limit of 100 pre-authorizations; 100 pre-authorizations have already been successfully processed today; billing coordinator is logged in; patient with valid insurance selects a new radiology test requiring pre-authorization.
Steps:
 1. Navigate to the patient's scheduling page.

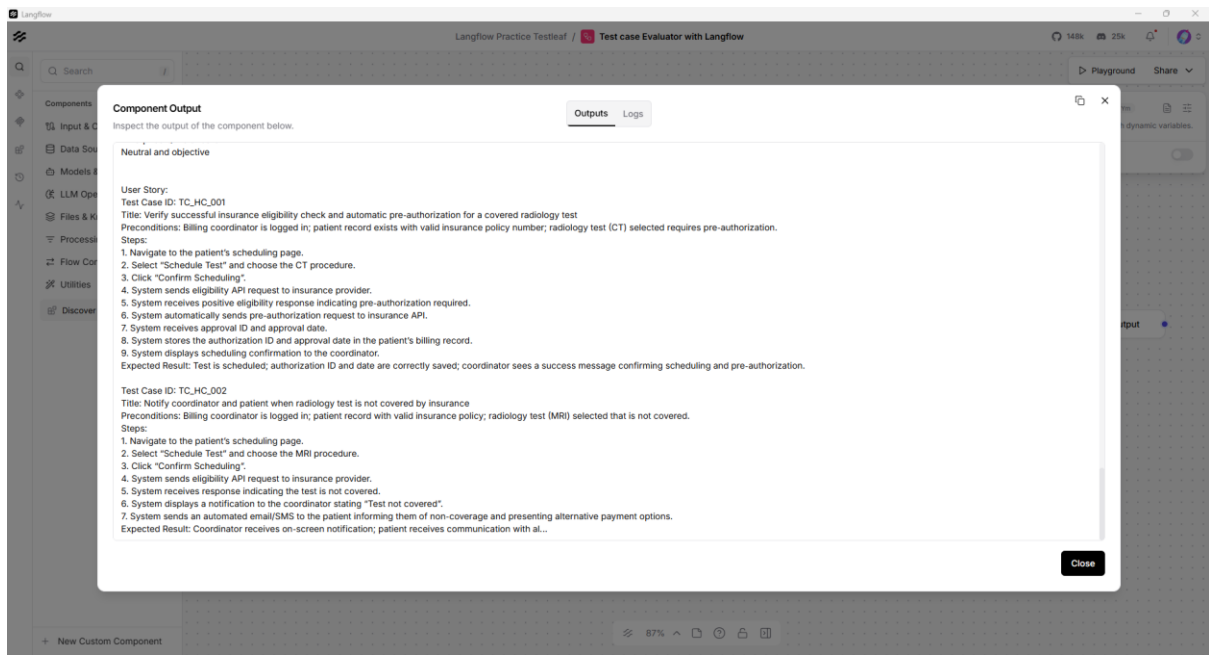


Test case Evaluator Prompt Template:

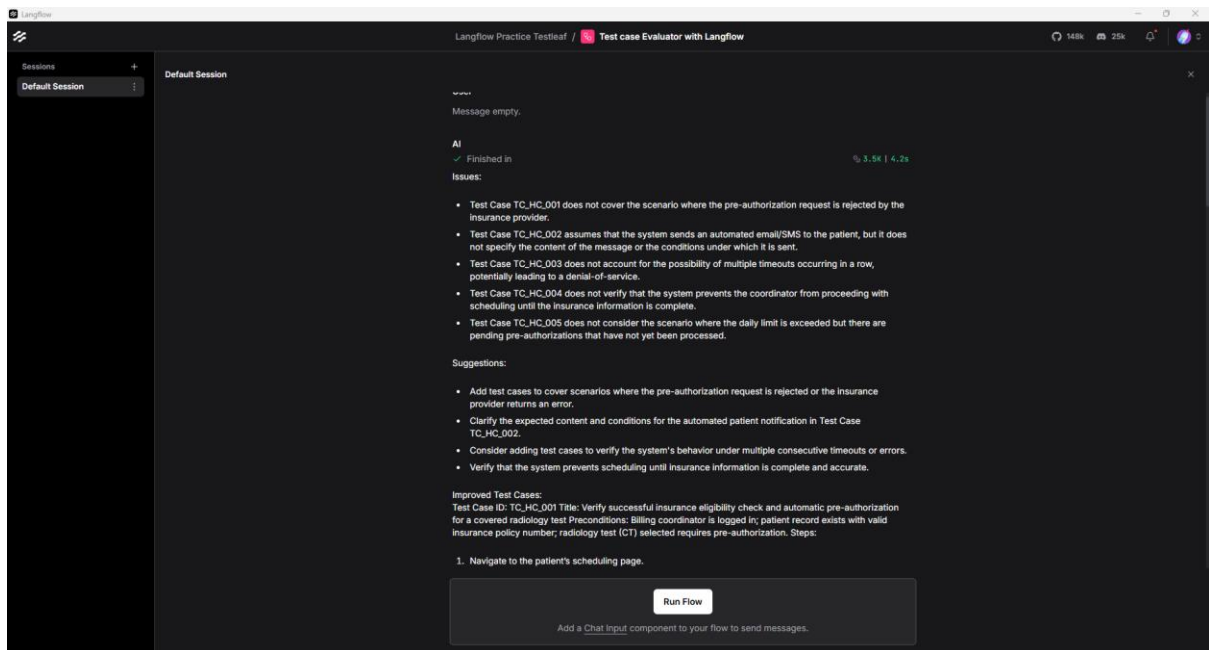








Test case Evaluator Output:



Langflow

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Sessions +

Default Session

Default Session

1. Navigate to the patient's scheduling page.

2. Select "Schedule Test" and choose the CT procedure.

3. Click "Confirm Scheduling".

4. System sends eligibility API request to insurance provider.

5. System receives positive eligibility response indicating pre-authorization required.

6. System automatically sends pre-authorization request to insurance API.

7. System receives approval ID and approval date.

8. System stores the authorization ID and approval date in the patient's billing record.

9. System displays scheduling confirmation to the coordinator.

Expected Result: Test is scheduled; authorization ID and date are correctly saved; coordinator sees a success message confirming scheduling and pre-authorization.

Test Case ID: TC_HC_002 Title: Notify coordinator and patient when radiology test is not covered by insurance

Preconditions: Billing coordinator is logged in; patient record with valid insurance policy; radiology test (MRI) selected that is not covered. Steps:

1. Navigate to the patient's scheduling page.

2. Select "Schedule Test" and choose the MRI procedure.

3. Click "Confirm Scheduling".

4. System sends eligibility API request to insurance provider.

5. System receives response indicating the test is not covered.

6. System displays a notification to the coordinator stating "Test not covered".

7. System sends an automated email/SMS to the patient informing them of non-coverage, including the following information:

• Test name

• Reason for non-coverage

• Alternative payment options

Expected Result: Coordinator receives on-screen notification; patient receives communication with alternative payment options; test is not scheduled.

Run Flow

Add a Chat Input component to your flow to send messages.

Langflow

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Default Session

Default Session

Test Case ID: TC_HC_003 Title: Handle insurance API timeout during eligibility verification

Preconditions: Billing coordinator is logged in; patient record with valid insurance policy; radiology test (Ultrasound) selected. Steps:

1. Navigate to the patient's scheduling page.

2. Select "Schedule Test" and choose the Ultrasound procedure.

3. Click "Confirm Scheduling".

4. System sends eligibility API request to insurance provider.

5. Insurance API does not respond within the configured timeout period.

6. System logs the timeout error.

7. System displays an error message to the coordinator: "Unable to verify insurance at this time. Please try again later."

8. Coordinator chooses "Retry" or "Cancel".

Expected Result: System gracefully handles timeout, logs the error, and provides the coordinator with retry/cancel options without crashing.

Test Case ID: TC_HC_004 Title: Validate error handling for missing insurance information in patient record

Preconditions: Billing coordinator is logged in; patient record exists but insurance policy number field is empty. Steps:

1. Navigate to the patient's scheduling page.

2. Select "Schedule Test" and choose any radiology procedure (e.g., CT).

3. Click "Confirm Scheduling".

4. System checks patient record for required insurance fields.

5. System detects missing insurance policy number.

6. System displays validation error: "Insurance information incomplete. Please update patient record before scheduling."

Expected Result: Scheduling is blocked; coordinator receives clear validation error; no API call is made; coordinator cannot proceed until insurance information is complete.

Test Case ID: TC_HC_005 Title: Verify system behavior when maximum daily pre-authorization limit is reached

Preconditions: System configuration sets a daily limit of 100 pre-authorizations; 100 pre-authorizations have already been successfully processed today; billing coordinator is logged in; patient with valid insurance

Run Flow

Add a Chat Input component to your flow to send messages.

Langflow

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Sessions

Default Session

Default Session

Test Case ID: TC_HC_005 Title: Verify system behavior when maximum daily pre-authorization limit is reached
Preconditions: System configuration sets a daily limit of 100 pre-authorizations; 100 pre-authorizations have already been successfully processed today; billing coordinator is logged in; patient with valid insurance selects a new radiology test requiring pre-authorization. Steps:

1. Navigate to the patient's scheduling page.
2. Select "Schedule Test" and choose a CT procedure that requires pre-authorization.
3. Click "Confirm Scheduling".
4. System sends eligibility API request and receives response indicating pre-authorization required.
5. System checks current daily pre-authorization count.
6. System determines the daily limit has been reached.
7. System displays error message: "Daily pre-authorization limit reached. Please try again tomorrow or contact support."

Expected Result: System prevents additional pre-authorization request, informs coordinator of the limit, and does not exceed configured daily threshold.

Test Case ID: TC_HC_006 Title: Verify system behavior when pre-authorization request is rejected by insurance provider
Preconditions: Billing coordinator is logged in; patient record exists with valid insurance policy number; radiology test (CT) selected requires pre-authorization. Steps:

1. Navigate to the patient's scheduling page.
2. Select "Schedule Test" and choose the CT procedure.
3. Click "Confirm Scheduling".
4. System sends eligibility API request to insurance provider.
5. System receives positive eligibility response indicating pre-authorization required.
6. System automatically sends pre-authorization request to insurance API.
7. Insurance provider rejects pre-authorization request.

Expected Result: System displays an error message to the coordinator indicating that the pre-authorization request was rejected; test is not scheduled.

Test Case ID: TC_HC_007 Title: Verify system behavior under multiple consecutive timeouts
Preconditions: Billing coordinator is logged in; patient record with valid insurance policy; radiology test (Ultrasound) selected. Steps:

Run Flow

Add a Chat Input component to your flow to send messages.

Langflow

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Sessions

Default Session

Default Session

Expected Result: System prevents additional pre-authorization request, informs coordinator of the limit, and does not exceed configured daily threshold.

Test Case ID: TC_HC_006 Title: Verify system behavior when pre-authorization request is rejected by insurance provider
Preconditions: Billing coordinator is logged in; patient record exists with valid insurance policy number; radiology test (CT) selected requires pre-authorization. Steps:

1. Navigate to the patient's scheduling page.
2. Select "Schedule Test" and choose the CT procedure.
3. Click "Confirm Scheduling".
4. System sends eligibility API request to insurance provider.
5. System receives positive eligibility response indicating pre-authorization required.
6. System automatically sends pre-authorization request to insurance API.
7. Insurance provider rejects pre-authorization request.

Expected Result: System displays an error message to the coordinator indicating that the pre-authorization request was rejected; test is not scheduled.

Test Case ID: TC_HC_007 Title: Verify system behavior under multiple consecutive timeouts
Preconditions: Billing coordinator is logged in; patient record with valid insurance policy; radiology test (Ultrasound) selected. Steps:

1. Navigate to the patient's scheduling page.
2. Select "Schedule Test" and choose the Ultrasound procedure.
3. Click "Confirm Scheduling".
4. System sends eligibility API request to insurance provider.
5. Insurance API does not respond within the configured timeout period (timeout 1).
6. Coordinator retries.
7. Insurance API does not respond within the configured timeout period (timeout 2).
8. Coordinator retries.

Expected Result: System handles multiple consecutive timeouts without crashing; coordinator is provided with retry/cancel options; system logs each timeout error.

Run Flow

Add a Chat Input component to your flow to send messages.